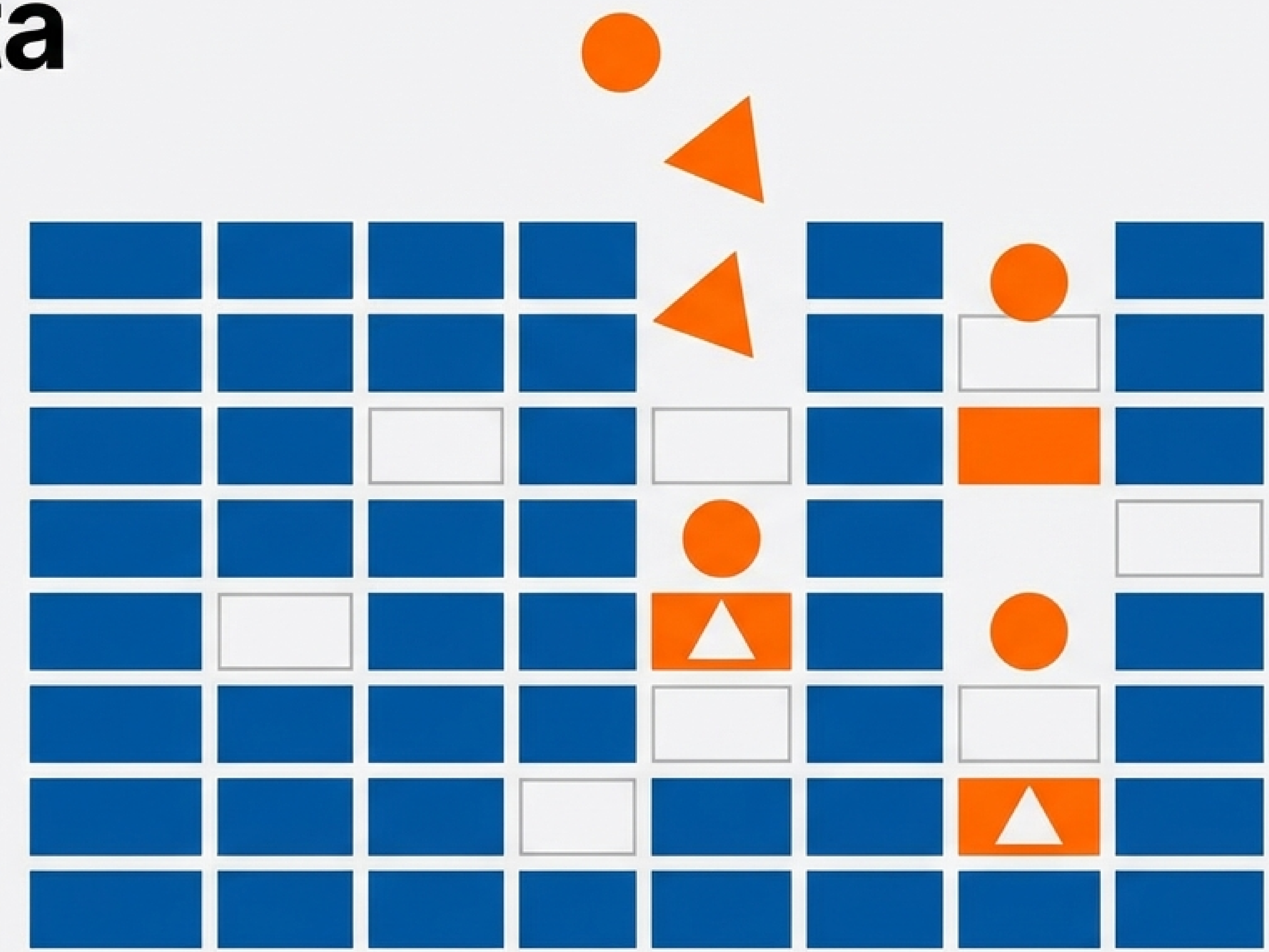
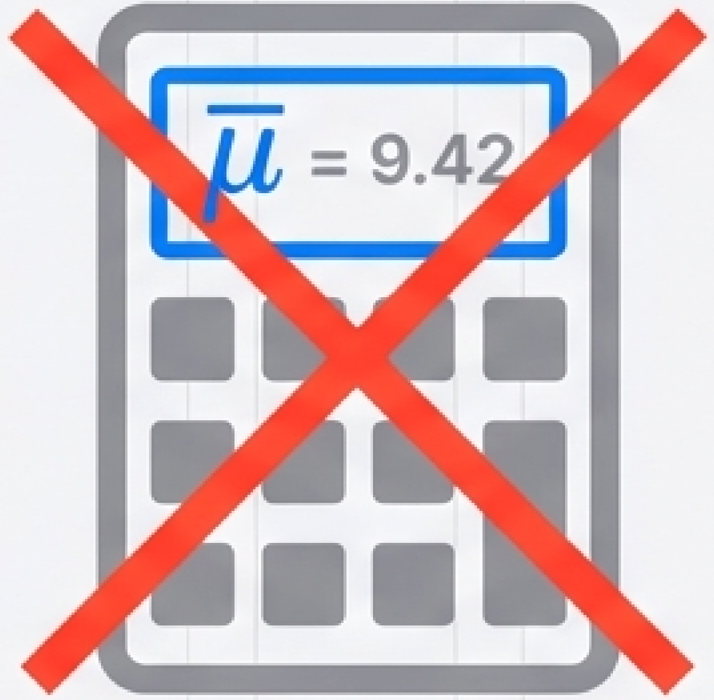
Mastering Missing Categorical Data Imputation

Strategies for handling non-numerical gaps in datasets using Pandas and Scikit-Learn.

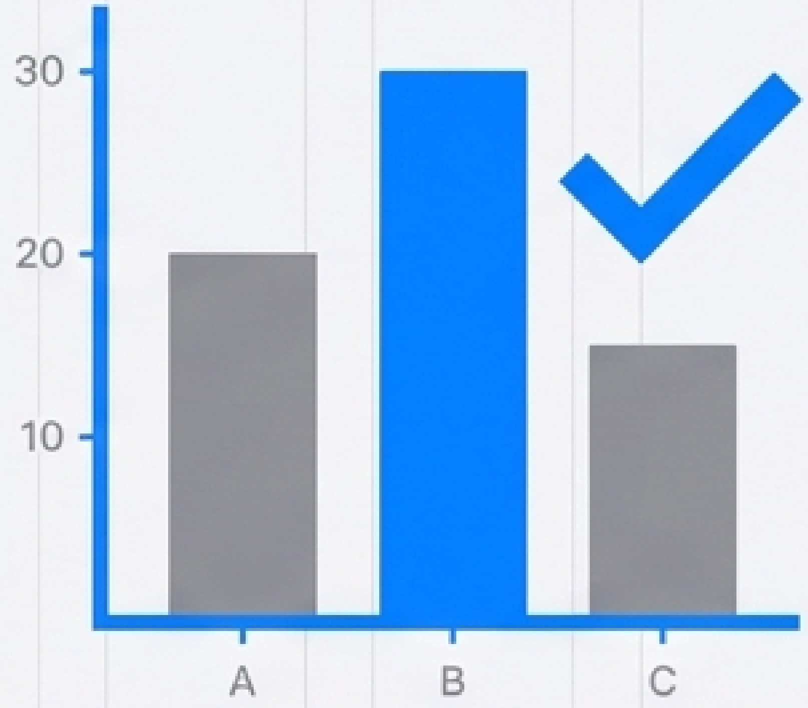


Case Study: Housing Dataset (Advanced Regression)

Why Mean and Median Don't Apply Here



Numerical Data: Mean/Median OK



Categorical Data: Frequency (Mode) or New Labels

In numerical datasets, missing values (<5%) are often replaced by the mean or median. However, you cannot calculate the average of categorical labels like 'City' or 'Colour'.

The Solution Space:

1. **Frequency-based:** Relying on the Mode (Most Frequent).
2. **Frequency-based:** Relying on the Mode (Most Frequent).
2. **Label-based:** Creating a new category explicitly for missing values.

The Two Primary Strategies

Strategy 1:
Most Frequent Imputation

Replaces NaN with the most common category (e.g., 'Mumbai').

Assumes data is Missing Completely At Random (MCAR).

Strategy 2:
Missing Category Imputation

JetBrains Mono

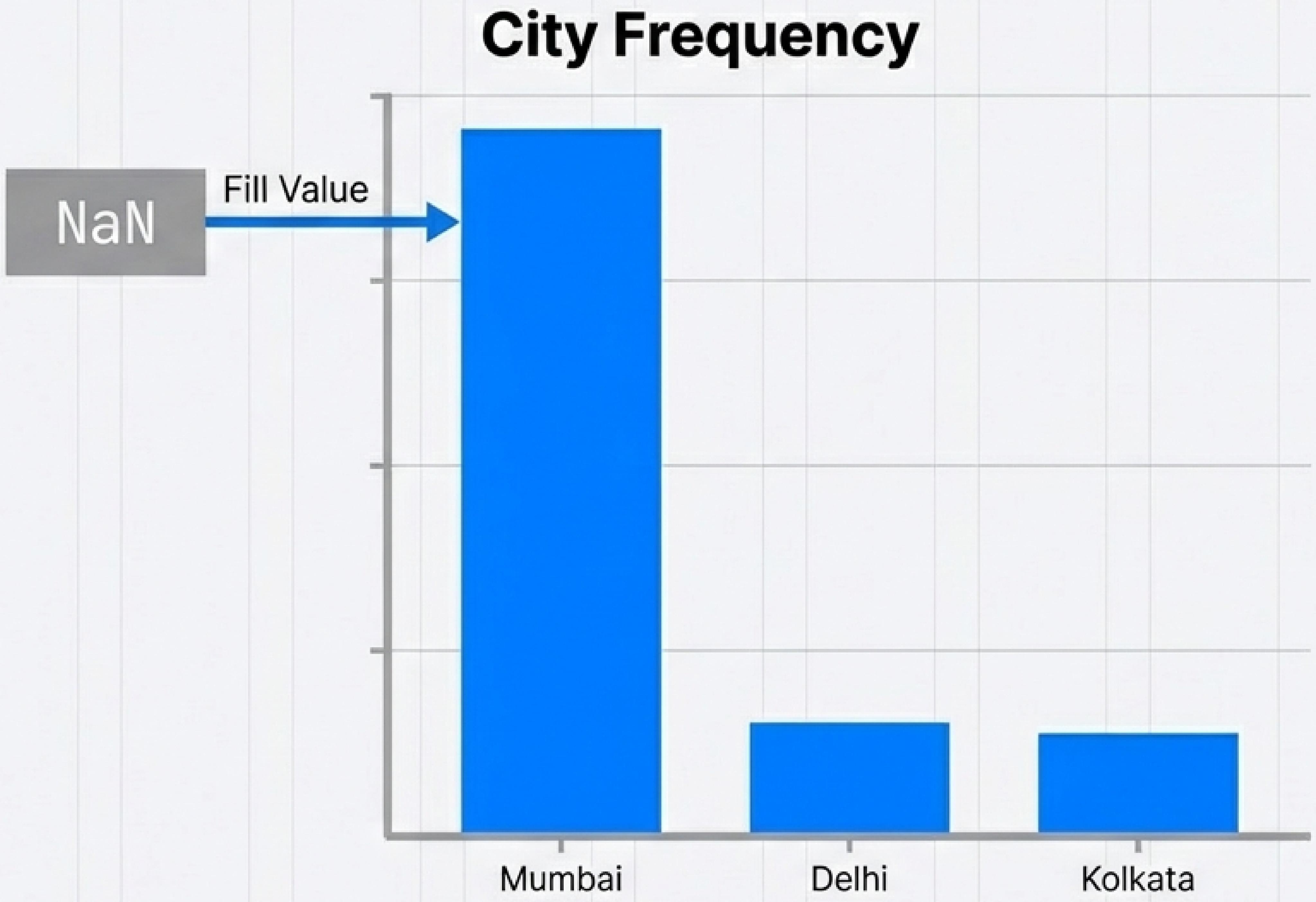
Treats NaN as a valid signal. Replaces missing values with a string literal.

Best for data Missing Not At Random (MNAR) or high missingness.

Strategy 1: Most Frequent Imputation (Mode)

The Prerequisites:

1. Low Missingness (< 5% of data).
2. Randomness (MCAR).
3. **The Dominance Rule:** One category must be significantly more frequent than others.



Validation: When Mode Imputation Works

Case Study: Garage Quality (~5% Missing). Dominant Category: 'TA' (Typical/Average).

Result: The curves overlap perfectly. The data structure remains intact because the missing values were few and the mode was dominant.

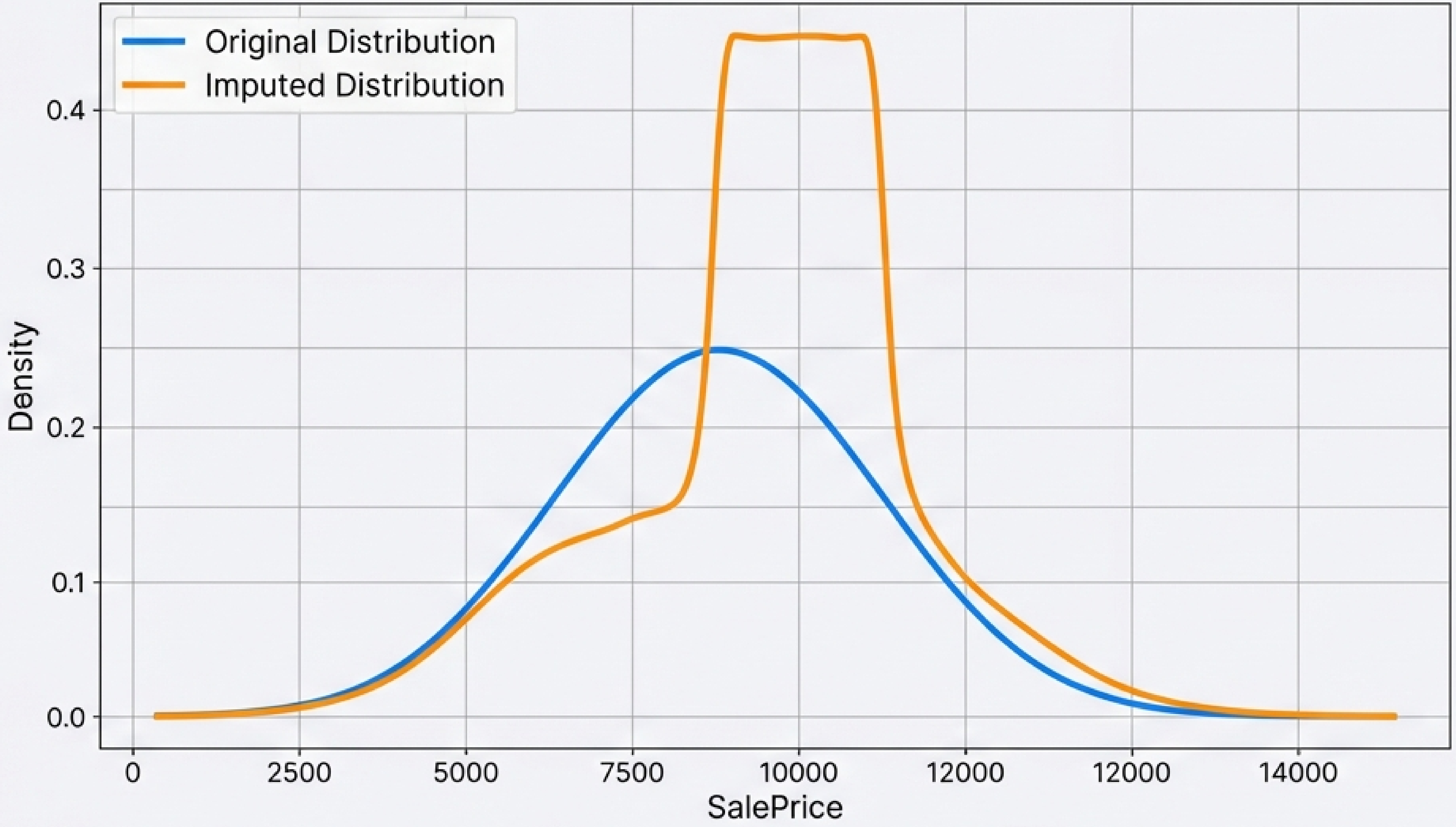
The Risk: When Mode Imputation Distorts Data

Case Study: Fireplace Quality (~50% Missing). Categories are balanced.

DISTORTION DETECTED

By forcing 50% of the data into the 'Good' category, we have artificially biased the dataset.

Do not use Mode imputation here.



Implementing Most Frequent Imputation

Pandas (Quick Analysis)

```
df['GarageQual'].fillna(df['GarageQual'].
mode()[0], inplace=True)
```

Scikit-Learn (Production Pipelines)

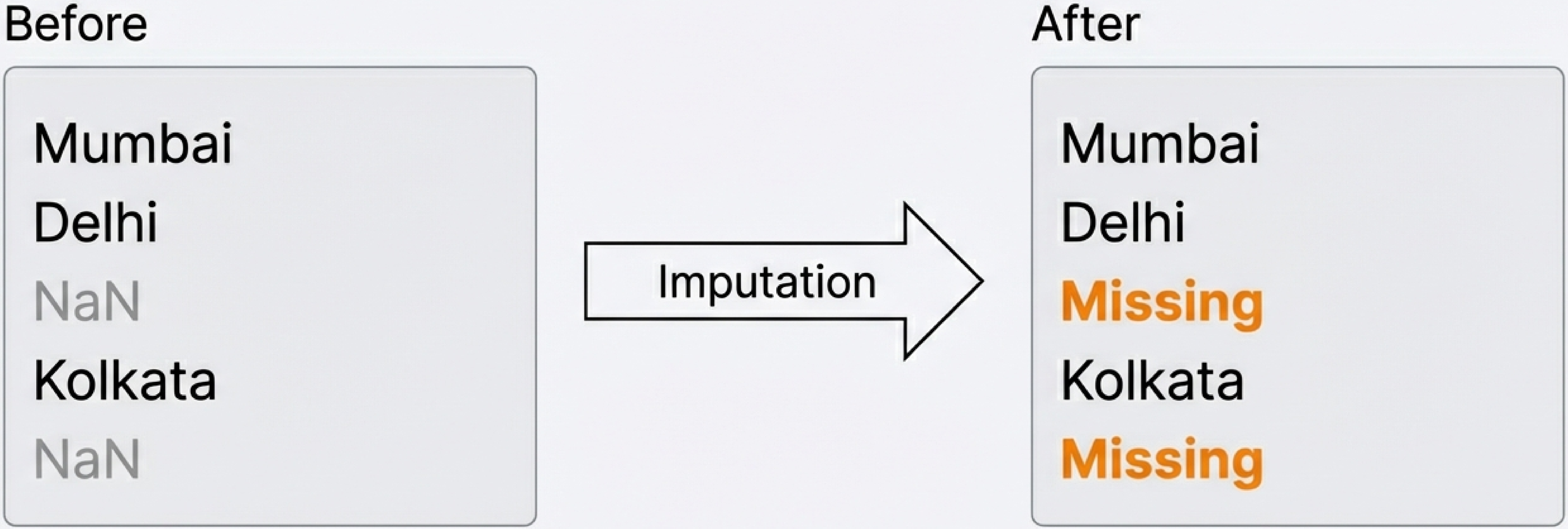
```
from sklearn.impute import SimpleImputer

imputer = SimpleImputer(strategy=
'most_frequent')
df['GarageQual'] = imputer.fit_transform(
df[['GarageQual']])
```

Pros	Cons
• Easy to implement; Compatible with linear models.	• Distorts distribution if missingness is high; • Ignores correlations.

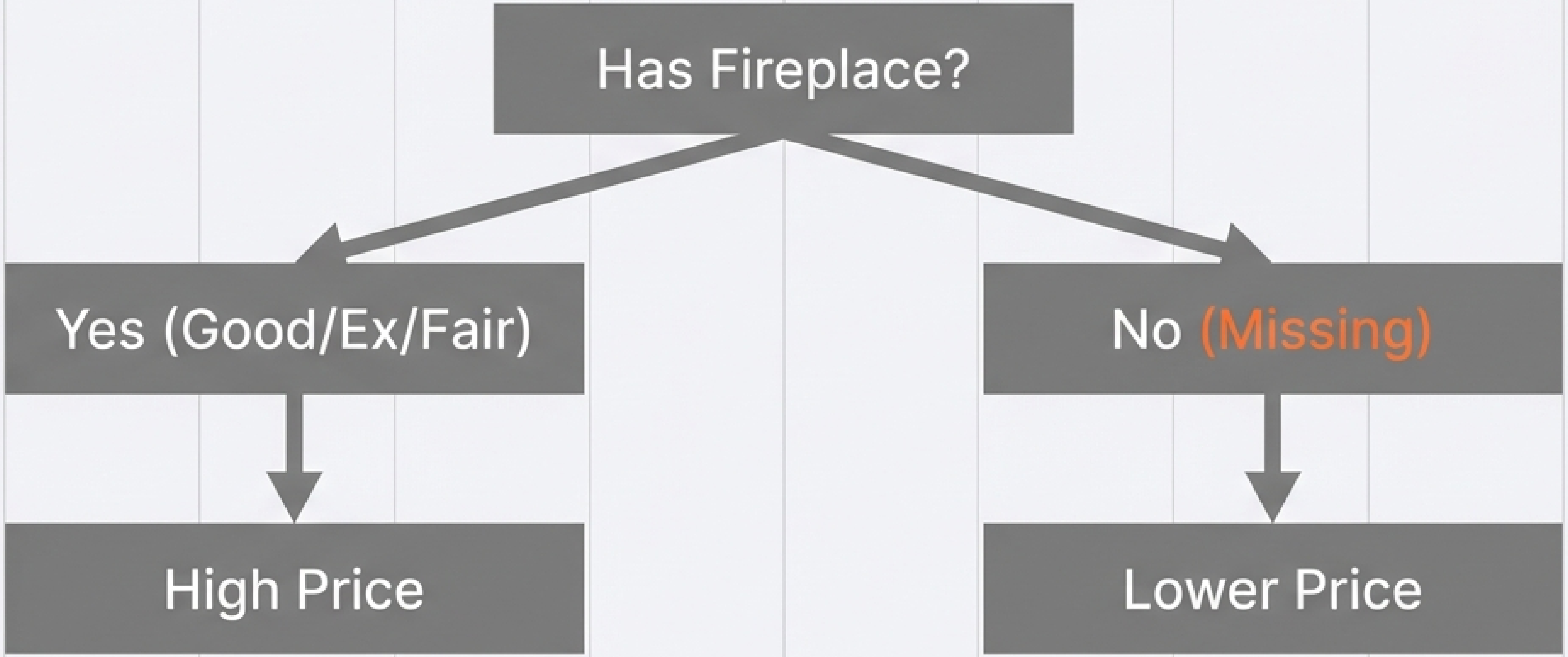
Strategy 2: Missing Category Imputation

Instead of guessing, we explicitly label the unknown. Ideally used when missingness > 10% or data is MNAR.



Why Explicit Labelling Works

Case Study: Fireplace Quality (50% Missing).



By labeling “NaN” as “Missing”, the algorithm treats the absence of information as a feature in itself. It learns that houses with “Missing” fireplace quality constitute a distinct price tier.

Implementing Missing Category Imputation

Pandas

```
df['FireplaceQu'].fillna('Missing',
inplace=True)
```

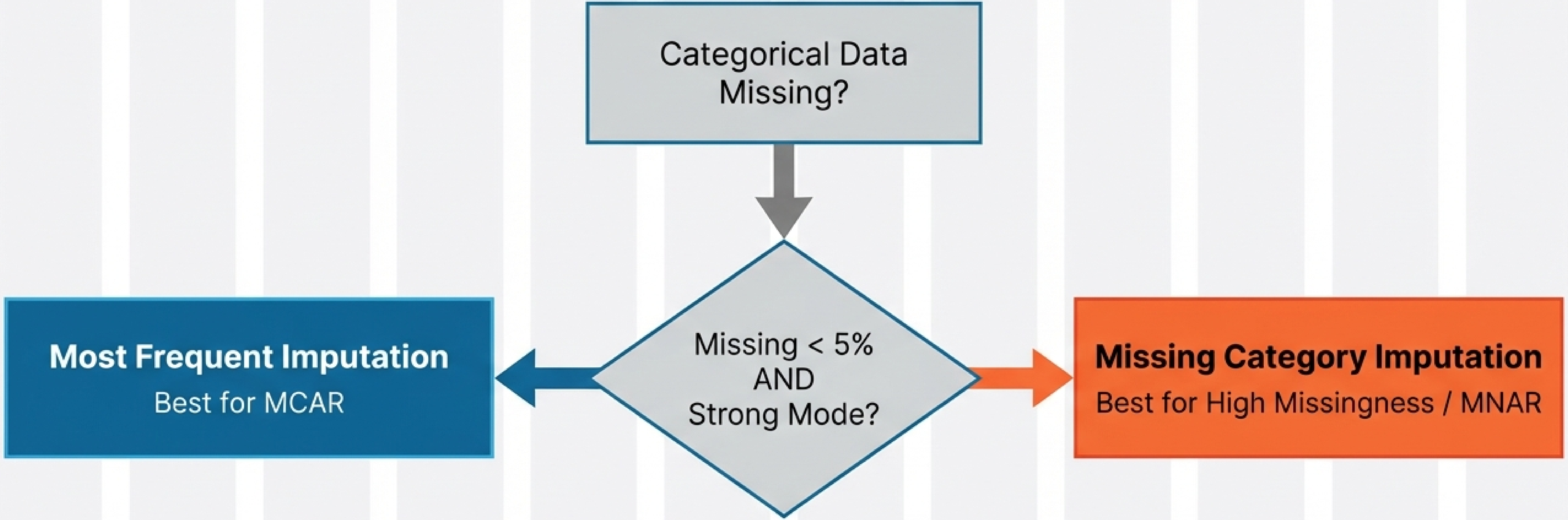
Scikit-Learn

```
imputer = SimpleImputer(strategy='constant',
fill_value='Missing')

df['FireplaceQu'] = imputer.fit_transform(df
[['FireplaceQu']])
```

Pros	Cons
Preserves distribution of existing categories; Prevents artificial bias.	Increases cardinality (adds a feature); Creates a category that may not physically exist.

Selecting the Right Strategy



If the categories are balanced (no clear mode), avoid **Most Frequent Imputation** even if missingness is low.

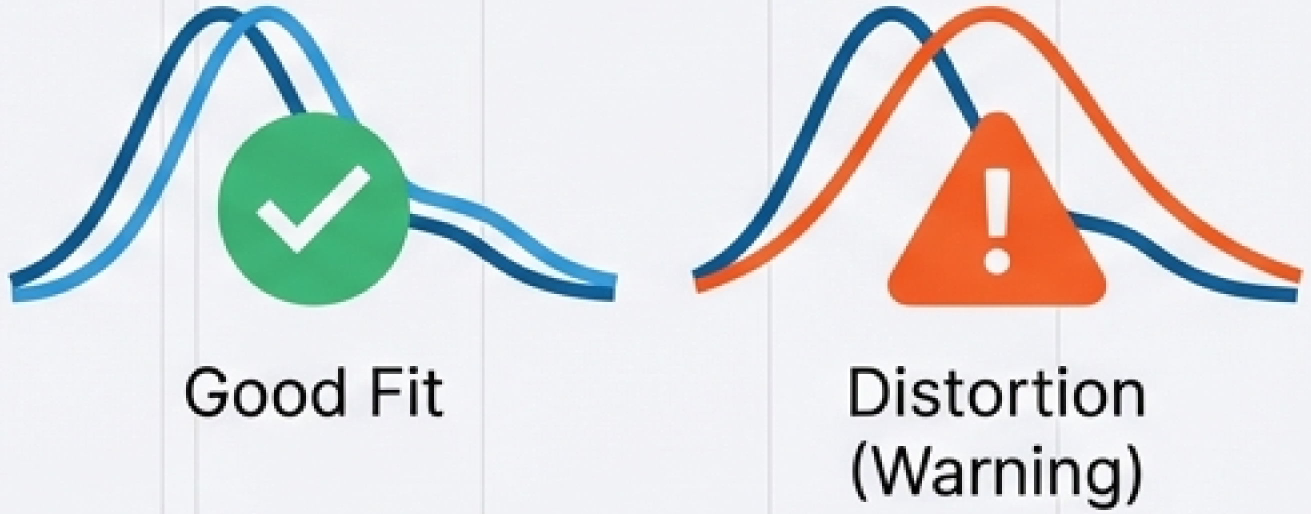
The Golden Rule: Validate with PDFs

Don’t just run code—analyse the impact on distribution.

1. Plot the Probability Density Function (PDF) of the target variable against the **Original** category.

2. Plot the PDF against the **Imputed** category.

3. **Compare**: If the curves diverge significantly, your imputation method is distorting reality.



Beyond Simple Imputation

While Mode and Constant imputation cover most categorical use cases, complex scenarios may require advanced techniques.

COMING SOON ↗

- **Random Imputation:** Sampling random values from the distribution to preserve variance.
- **Multivariate Imputation (KNN): Using other features** (e.g., House Price, Location) to predict the missing category intelligently.

Master the fundamentals of distribution analysis first.